

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

EMEM gives an observation a lookup identity and a content-addressed record. Agent A passes the reference. Agent B resolves the same record and checks it before continuing.

Earth data stays in the data layer. Its reference enters the reasoning layer.

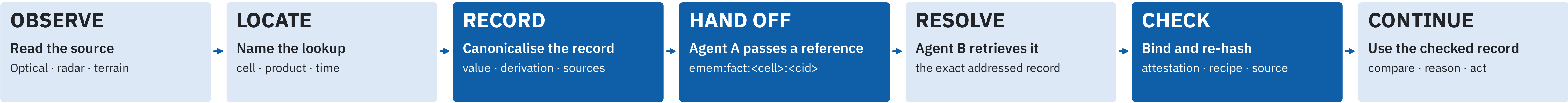
Jaya Kumari · Avijeet Singh · Vortex AI · avijeet@vortex.ai
Agentic AI for Earth Observation · BIFOLD and ESA Φ-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)

1 How does Earth data enter an agent workflow?

Keep the source data upstream. Pass its reference.

Locate an observation, record what was read, and carry its content address into the next agent's reasoning.

EARTH DATA



Lookup identity: where / product / time

Content address: BLAKE3 of the canonical record

CONTROLLED HANDOFF TEST: ACTED ON CORRUPTED EVIDENCE

prose
254 / 276

JSON
164 / 276

retrieved text
220 / 276

opaque id
154 / 288

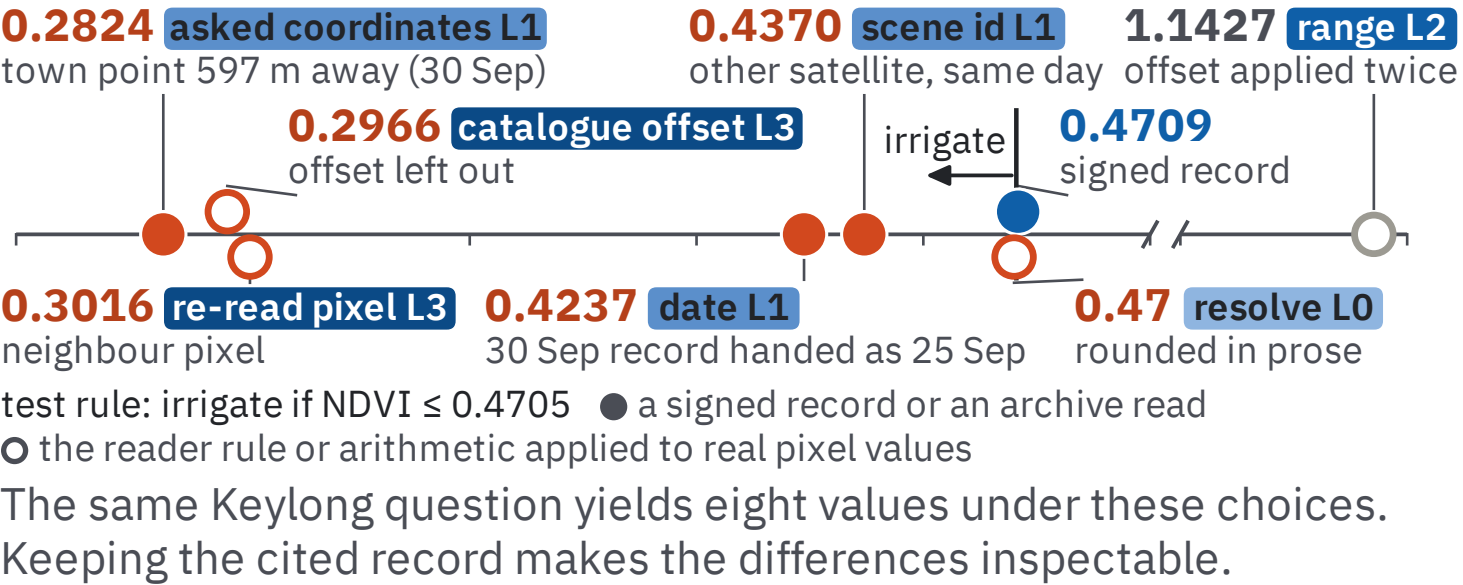
checked reference
0 / 300

In the controlled handoff test, prose led agents to act on 254 of 276 corruptions; a reference with explicit checking instructions, 0 of 300.

2 What can change between agents?

One question, several records

Scene, pixel, offset, rounding and stale state can change the value passed on.



3 Where did the evidence change?

A changed value needs a cause

Keep the scene, pixel, model and time attached so a change can be investigated.

$$\Delta Z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

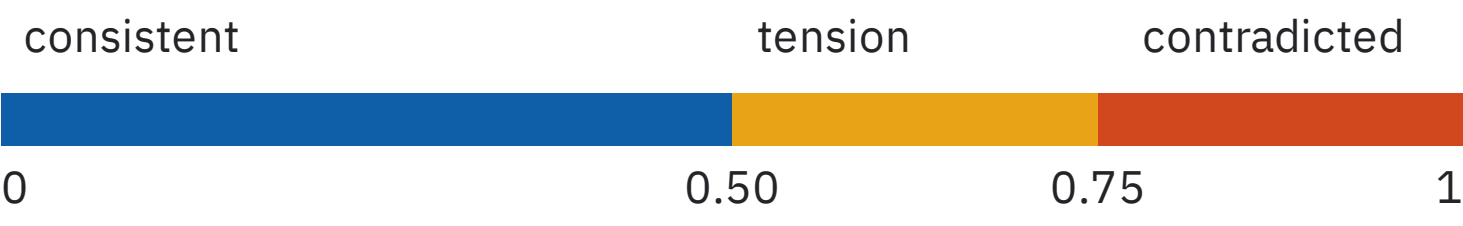
World, instrument, alignment, model and noise. This is an attribution model; the numeric split remains open work.

IMPLEMENTED · DRIFT-ANCHOR SCORE

$$u = |x - a| / (3\sigma)$$

$$s = u / (1 + u)$$

x: output · a: anchor · σ: anchor uncertainty



Tests pin 3σ → 0.50 and 9σ → 0.75.

Missing or invalid σ: exact agreement required.

The implemented anchor score measures disagreement, not its cause. SAT-042 exercises it in a reference harness; live device-to-anchor wiring remains next work.

Memory = observations + edges

$$M = (O^*, E^*)$$

O* stores observations; E* stores typed temporal relations. Append new records and keep earlier ones.

E = (subject, predicate, object, t_{from}, t_{to})

Relations can say supersedes or disagrees_with, with a validity interval.

Lookup: (cell, band, observation time)
cid(O) = BLAKE3(CBOR_{canonical}(O))

recall(M, a, b | t*, τ)

At address a and band b, bound observation time by t* and what memory knew by signing time τ.

Model outputs are records too

Satellite input → encoder → vector record → content address

Prithvi 1,024 values

checkpoint digest inside the record

TESSERA 128 values

product year 2024; no checkpoint digest

Archived vectors: Prithvi names a checkpoint digest; TESSERA names a product year. These encoders are retired on the deployment; stored records remain addressable.

Missing data stays explicit

An absence record carries a reason and source. Berlin's CHIRPS record says outside the product's latitude range; it supplies no rainfall value.

Where emem fits. STAC finds assets; openEO runs processing; PROV and C2PA carry provenance. EMEM adds the observation reference an agent can resolve and re-check.

11 Use emem. Carry the evidence forward.

Connect a plugin, a workflow or your own code. Public emem reads need no API key; the source is open.

Measured separately: the same record and value through 11 client paths; receipt signatures checked on 9 (30 Sep 2026).

The routes below are plugins, connectors, packages and listings. One evidence protocol, multiple agent runtimes. Setup and compatibility notes are behind CONNECT.

CONNECT Copy a token, choose your tool, run the checks. vortex-ai.github.io/esa_postexr/use/

Attested observation

→ MCP / A2A

→ Agent host / client

→ Next agent / application

ChatGPT @emem plugin Enable emem; paste a token.	Claude Code plugin / MCP Resolve from chat or terminal.	Visual Studio Code MCP connector Copilot agent mode: @mcp emem	Diffy Marketplace plugin Referent Lock / EU DR workflows	Salesforce MuleSoft Exchange listing Open Vortex AI MCP Server
---	--	---	---	---

MORE CLIENTS

Cursor
Gemini CLI

PROTOCOLS

MCP · A2A
REST / OpenAPI

DEVELOPER

Python · TypeScript
Docker · LangGraph memory
GitHub source

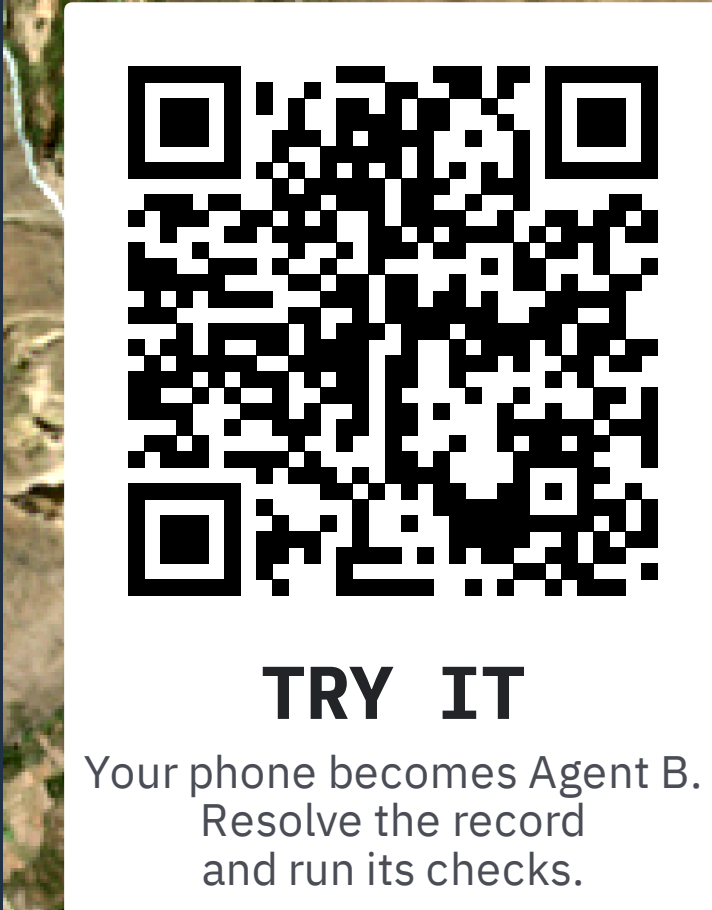
FRAMEWORK EXAMPLES

LangChain · LlamaIndex
AutoGen · CrewAI · Mastra
Agno (see setup notes)

DISCOVERY / MIRRORS

GitHub MCP · MCP Registry
Glama · Smithery · ClawHub
Context7 · Hugging Face (older mirror)

Implementation: emem 8e9b401. Measurements: 29 Sep to 1 Oct 2026. · The linked methods connect this poster's figures to their records, source data and reproducible checks. · Related work: Sigstore, RFC 9162, SCITT (RFC 9943), ARC (arXiv 2607.25066). · Sentinel-2 Products Specification (ESA) · Copernicus DEM GLO-30/90 · Hansen et al. 2013, GFC v1.13 · JRC GFC2020 V3/V4, TMF · ESA CCI Biomass v7 · BLAKE3 · RFC 8032 · RFC 6962/9162 · STAC 1.1 · openEO 1.3 · W3C PROV-DM · C2PA 2.4 · MCP 2025-11-25 · A2A 1.0 · Perez et al., ICLR 2025 · Munir et al. 2026 (2604.24919) · Cemri et al. 2025, MAST (2503.13657) · Townshend et al. 1992 · GeoGuard (NASA-IMPACT) · Prithvi-EO-2.0 · TESSERA (2506.20380)



TRY IT

Your phone becomes Agent B.
Resolve the record
and run its checks.

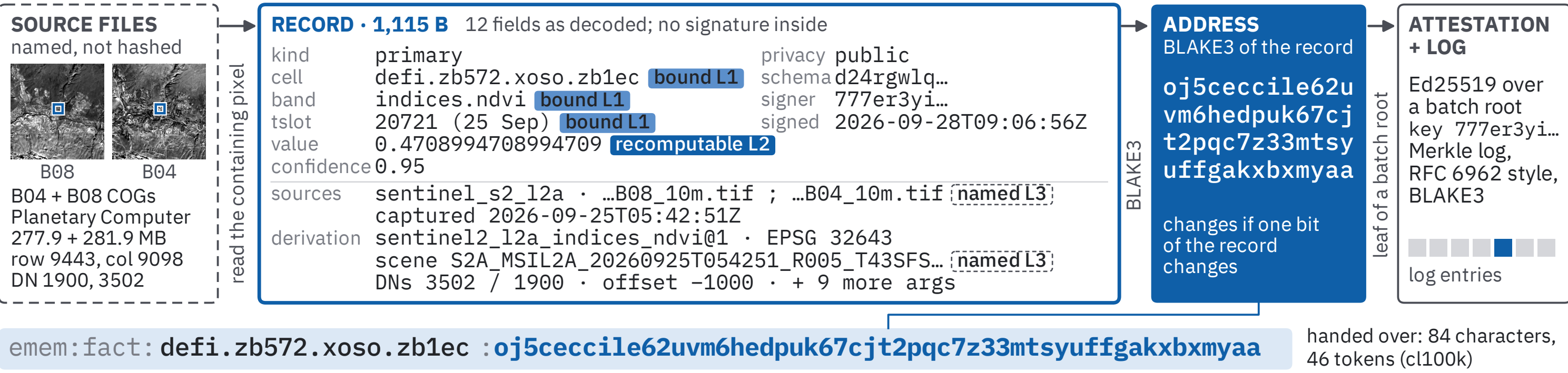
The record this board follows: NDVI 0.4709
Sentinel-2A L2A · Keylong, Lahaul, India · 25 Sep 2026
true colour B04/B03/B02 from signed 10 m grids; linear 1 to 99 % stretch, y 1/1.35

4 What exactly is handed over?

A compact reference opens the observation record

Resolve the value, place, time, derivation and source references from the record's address.

INSPECT The 1,115 bytes behind this example.



emem:fact: defi. zb572. xoso. zb1ec :oj5ceccile62uvm6hedpuk67cj2tpqc7z33mtsuyffgakxbxmyaa
BLAKE3 identifies the exact 1,115-byte record. A batch attestation covers its address. The record names the source files and read location; source checking continues by re-reading them.

5 Which changes can the receiver detect?

Checks can continue back to the source

Resolve → Re-hash → Bind → Recompute → Re-read

agents as B: three Claude models pooled, 1 Oct 2026	prose	JSON	RAG	opaque id	EMEM, full check	first check that refuses
■ B acted on corrupted evidence	■ unaffected	■ refused	■ not applicable			
■ M2 seen in production	■ the only check that stops it	■ flips the decision				
G0 nothing altered: accepted everywhere, never refused						
M1 value	stated value moved by 1 ULP	12/12	12/12	12/12	9/12	C L0
M2	stated value rounded to "0.47"	6/12	6/12	11/12	0/12	C L0
M8	value 0.45, re-encoded and re-hashed	12/12	6/12	12/12	6/12	F L0
M4 cell	record cited for another cell	7/12	0/12	6/12	6/12	E L1
M9	cell changed inside the record, re-hashed	7/12	11/12	6/12	6/12	F L0
M10 time	tslot changed to look current, re-hashed	12/12	6/12	6/12	6/12	F L0
M6 band	a record for another band handed over	12/12	0/12	6/12	2/12	E L1
M11 source	source scene id changed, re-hashed	12/12	12/12	12/12	12/12	F L0
M12 derivation	offset changed, value recomputed, re-hashed	12/12	11/12	12/12	8/12	F L0
M14	value (0.46) disagrees with the signed DNs	12/12	6/12	12/12	6/12	H L2
M5 stale / current	an older record handed over as current	12/12	1/12	6/12	1/12	E L1
M16	a second signed version shown only to B	12/12	6/12	6/12	6/12	G L0
M3 signature / id	1 ULP changed inside the served bytes	12/12	12/12	12/12	12/12	D L0
M7	token miscopied by one character	n/a	n/a	n/a	0/12	C L0
M13	value 0.45 under the forger's own key	12/12	6/12	12/12	6/12	F L0
M15 source pixel	DNs read from the pixel 10 m south	12/12	12/12	12/12	12/12	I L3
M17 entity	same record; A meant another entity (out of scope)	not run	not run	not run	not run	none

B acts on corrupted evidence

C resolve L0 D hash L0 E binding L1 F signature L0 G log L0 H recompute L2 I re-read L3 D to I are cumulative: each adds one check

Bars: agents' false acceptance, k of n per cell, n 12 (6 Haiku 4.5, 5 Sonnet 5.5, 1 Opus 5.5 runs); squares: the deterministic ceiling. Totals pool 23 items, 1 Oct 2026.

Three further signer errors (offset, same-day scene, unit) pass checks D to I; metadata checks catch them.

Deeper checks expose different corruptions. Checked-reference agents made the expected decision on 71 of 72 genuine controls; one was refused (pooled Claude).

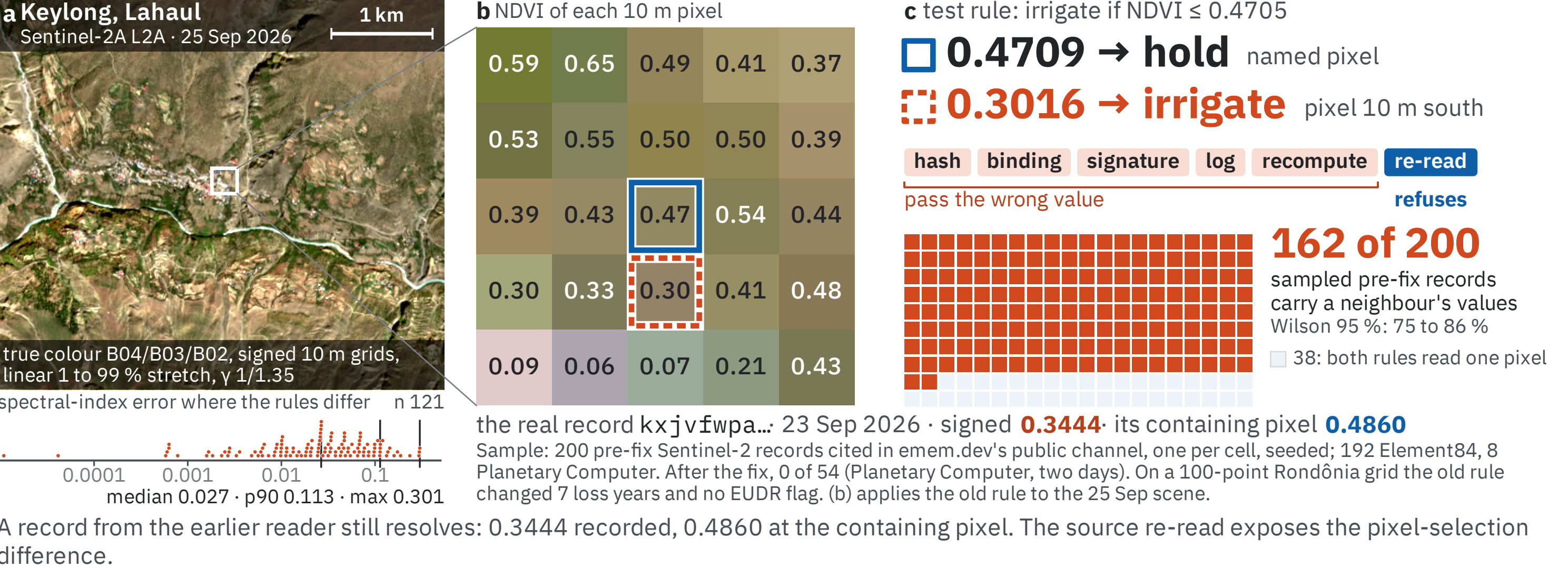
Agent bars: 12 runs per item, across 23 items. JSON and an opaque id caught some changes from their fields. In the live pre-fix record case, agents acted on 4 of 4 uninstructed, 2 of 4 with resolve-and-bind instructions, and 0 of 4 with a source re-read.

6 What does returning to the source add?

A valid record can preserve a wrong source read

$$NDVI = (DN_8 - DN_4) / (DN_8 + DN_4 + 2o), o = -1000$$

The recorded product offset o is applied to both bands before the ratio.



A record from the earlier reader still resolves: 0.3444 recorded, 0.4860 at the containing pixel. The source re-read exposes the pixel-selection difference.

7 What has the receiver established?

Know what was checked

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Checks establish properties of the cited record. A source re-read adds evidence; sensor accuracy remains inherited.

8 Can a later agent recover the earlier state?

Memory keeps what was cited

BENGALURU · SAME CELL AND BAND · 2026

signed 28 May	signed 11 Aug
918.0 m	915.07 m
Open-Meteo / DEM90 earlier CID	Copernicus DEM30 later CID

as of 15 Jun → 918.0 m

The next agent can recover the earlier citation.

Asked what memory knew on 15 Jun, recall returns 918.0 m. The later 915.07 m record reflects a provider change; it does not establish ground movement.

9 What does one Berlin location reveal?

One place, many observations

Central Berlin: optical, radar, elevation, temperature, water, forest and biomass. One lookup cell organises 15 products, preserving each record and native grid.



product	signed reading	valid time	grid	fact_cid
Sentinel-2C L2A	NIR 0.0475	27 Sep 2026	10 m	wtdq64...
Sentinel-2C L2A	NDVI 0.195	27 Sep 2026	10 m	y2k2ti...
Sentinel-1C RTC	VV -2.82 dB	28 Sep 2026	n/s	2x2enx...
GLO-30 DEM	36.6 m DSM	2021	30 m	sdu3gf...
MODIS MOD11A2	297.4 K day LST	29 Aug 2026	1 km	yqhihm...
ESA WorldCover	built-up (50)	2021	10 m	xicg6u...
ESA CCI Biomass	0 t/ha	2022	100 m	keebyc...
JRC GSW v1.4	0 % water	1984 to 2021	30 m	hohgux...
Hansen v1.13	0 % tree cover	2000	n/s	x4kpkj...
JRC GFC2020 V4	no forest	2020	10 m	oog2b4...
Copernicus CAMS	NO ₂ 12.3 µg/m³	30 Sep 2026	~11 km	jvwhwg...
Overture, not EO	0 buildings	Aug 2026	vector	czehax...
ISRIC SoilGrids	no data, null	30 Sep 2026	250 m	mkwinv...
CHIRPS v2.0	outside ±30°	30 Sep 2026	0.05°	bkaovt...
JRC TMF	outside ±30°	30 Sep 2026	30 m	w7t2hw...
NASA FIRMS	no fire in 24 h	30 Sep 2026	points	k2ykb2...

15 products, 16 signed facts, 4 signed absences, 10 m to about 11 km; read live 30 Sep 2026. Native grids remain distinct; n/s: grid not in source files. 16 records re-hash under emem.dev's key, 2 Oct 2026.

10 What else can an agent carry?

Beyond a single observation

EARTH DATA
raster → cube → raster
Fields and dated stacks; retain the scene.
COLLECT EVIDENCE
8 facts → one bundle handle
38 characters; members resolve separately.
REASONING CHECKPOINTS
located → routed → recalled → scored
4 stage addresses re-hashed offline.
FILES, IDENTITY AND EXECUTION
tree · entity · trace · track
Address chunks, subjects and recorded runs.

Eight facts resolve through one bundle; four checkpoint hashes reproduce. These object checks do not validate an agent's reasoning.

NEXT EXPERIMENTS

Measure each cause of change

Validate numeric attribution under controlled scene, pixel and encoder changes. Repeat checked handoffs between separately operated agent hosts.

